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This is Exam 2 of MAT1120. You must show all work when required in order to receive credit. This exam is worth 100 points (roughly one sixth of your total grade thus far). You may consult anything in your text or posted to ASULearn. You may not consult anyone other than me. If I suspect academic dishonesty then the burden of proof is on you to demonstrate the material live. I will be available in Friday office hours (9am EST to 9:45am EST in person and zoom, or 12pm to 1pm in person only) if you need clarification on a problem, else I will respond via email whenever I can. **No late exams will be accepted without prior approval given solely at my discretion.** Follow the directions of each problem. Review your submission after turning in the exam and re-upload anything that might be difficult to grade.

Part I - Techniques of Integration

Perform the following integrals. The technique of choice is up to you. Each integral is worth 6 points for a total of 60 points.

1) $\displaystyle\int \frac{e^x}{4+e^{2x}}\,dx$

2) $\displaystyle\int \tan(x)\ln(\pi\cos(x))\,dx$

3) $\displaystyle\int_1^{\infty} \frac{1}{1+x^2}\,dx$

4) $\displaystyle\int \sin^3(4x)\cos(4x)\,dx$

5) $\displaystyle\int \frac{x+1}{x^2-5x+6}\,dx$

6) $\displaystyle\int_3^{19} \frac{4}{\sqrt{x-3}}\,dx$

7) $\displaystyle\int x^2\ln(x^3)\,dx$

8) $\displaystyle\int x^3\ln(x^2)\,dx$

9) $\displaystyle\int e^{2x+3}\cos(2x+3)\,dx$

10) $\displaystyle\int \frac{x^3-2}{x^2+1}\,dx$

Part II - Numeric Approximations

For problems 11-13 let $f(x) = \frac{1}{\sqrt{2\pi}} e^{-0.5x^2}$ and round all answers to four decimal places of precision.

This section is worth a total of 20 points.

11) Approximate $\int_{-1}^{1} f(x)\ dx$ using the trapezoid rule with 8 partitions. Also compute the maximum error associated with your approximation.

12) Approximate $\int_{-1}^{1} f(x)\ dx$ using the midpoint rule with 8 partitions. Also compute the maximum error associated with your approximation.

13) Approximate $\int_{-1}^{1} f(x)\ dx$ using Simpson's rule with 8 partitions. Also compute the maximum error associated with your approximation. (There's no shenanigans here, $f(x)$ is always positive and we have an even number of partitions.)

It should be noted that the curve in question is **the bell curve** from statistics, and according to the empirical rule from statistics we should expect around 68% of the area under the curve to be found within 1 standard deviation of the mean. We define the standard deviation of the bell curve to be the distance between the mean and the inflection points of this curve... which happen to be at -1 and 1. All of this is to say your answers should hover around 0.68.

Part III - Differential Equations

14) Find the particular solution to the initial value problem. (5 points.)

$$\frac{dy}{dt} = 3t^2(1 + y^2) \qquad y(0) = \frac{\pi}{4}$$

15) Find the general solution to the given differential equation. Do not isolate y in your final solution, leave it as you find it post integration. (5 points.)

$$\frac{dy}{dt} = \frac{\cos(t)(y^2 - 1)}{2y^3}$$

16) A 120 liter tank of water initially has 0.5 kilograms of sugar dissolved into it. The Mad Mixer is pouring in sugar water at a rate of 5 liters per minute and each liter of his sugar water contains 2.3 kilograms of sugar. At the same time, our hero is draining the well mixed solution at a rate of 5 liters per minute in a desperate attempt to prevent too much sugar from building up in the tank. Find the particular solution to the differential equation being described here and predict when the sugar levels will reach a total of 120kg (10 points).

Bonus: Let $y(x) = Ce^{\frac{x}{2}}$. Verify that $y(x)$ is a solution to the infinite order differential equation

$$y(x) = \sum_{n=1}^{\infty} y^{(n)}(x)$$

Where $y^{(n)}(x)$ represents the nth derivative of $y(x)$. (Hint: the current materials will help.)

1) $\int \frac{e^x}{4+e^{2x}}dx$

$u = e^x \quad du = e^x dx$

$\int \frac{1}{4+u^2} du$

$\frac{1}{4}\int \frac{1}{1+\frac{u^2}{4}}$

$v = \frac{u}{2} \quad dv = \frac{1}{2}du$

$\frac{1}{2}\int \frac{1}{v^2+1} dv$

$\frac{1}{2}\tan^{-1}(v) + C$

$\frac{1}{2}\tan^{-1}\left(\frac{u}{2}\right)+C$

$\frac{1}{2}\tan^{-1}\left(\frac{e^x}{2}\right)+C$

2) $\int \tan(x)\ln(\pi\cos(x))\,dx$

$u = \ln(\pi\cos x)$

$du = -\tan x$

$\frac{du}{dx} = \frac{1}{\pi\cos x}\cdot\frac{d}{dx}(\pi\cos x)$

$\frac{-\pi\sin x}{\pi\cos x} = -\tan x$

$du = -\tan x\,dx \Rightarrow \tan x\,dx = -du$

$\int u(-du) = -\int u\,du$

$-\frac{1}{2}u^2 + C$

$-\frac{1}{2}\ln(\pi\cos(x))^2 + c$

3) $\int_1^\infty \frac{1}{1+x^2}dx$

$\lim_{b\to\infty} \tan^{-1}(x)\Big|_1^b$

$\tan^{-1}(x)\Big|_1^\infty = \tan^{-1}(\infty) - \tan^{-1}(1) = \frac{\pi}{4}$

$\frac{\pi}{2} - \frac{\pi}{4}$

4) $\int \sin^3(4x)\cos(4x)\,dx$

$u = 4x \quad du = 4dx$

$\frac{1}{4}\int \sin^3 u \cos u\,du$

$v = \sin u \quad dv = \cos u\,du$

$\frac{1}{4}\int v^3 dv$

$\frac{v^4}{16} + C$

$\frac{\sin^4 u}{16} + C$

$\frac{1}{16}\sin^4(4x) + c$

5) $\int \frac{x+1}{x^2-5x+6}dx$

$(x-2)(x-3)$

$\frac{x-1}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3}$

$\frac{x-1}{x-2} + \frac{x-1}{x-3}$

$x=2 \quad \frac{2+1}{2-3} = -3 \qquad x=3 \quad \frac{3+1}{3-2} = 4$

$\int\left(\frac{-3}{x-2} + \frac{4}{x-3}\right)dx$

$-3\ln|x-2| + 4\ln|x-3| + C$

6) $\int_3^{19} \frac{4}{\sqrt{x-3}}dx$

$\lim_{t\to 3+} \int_t^{19} 4(x-3)^{-\frac{1}{2}}dx$

$\int 4\frac{(x-3)^{\frac{1}{2}}}{\frac{1}{2}}$

$8\sqrt{x-3}$

$\lim_{t\to 3+} \left[8\sqrt{x-3}\right]_t^{19}$

$\lim_{t\to 3+} \left(8\sqrt{19-3} - 8\sqrt{t-3}\right)$

$8\sqrt{0}$

$8\cdot 4$

$32 - 0$

32

7) $\int x^2 \ln(x^3)\,dx$

$\int x^2 \cdot 3\ln(x)dx$

$3\int x^2 \ln(x)dx$

$u = \ln x \qquad dv = x^2dx$ u-v- subs

$du = \frac{1}{x}dx \qquad v = \frac{x^3}{3}$

$\ln(x)\frac{x^3}{3} - \int \frac{x^3}{3}\cdot\frac{1}{x}dx$

$\ln(x)\frac{x^3}{3} - \frac{1}{3}\int x^2 dx$

8) $\int x^3 \ln x^2 dx$

$\int x^3 \cdot 2\ln(x)\,dx$

$2\int x^3 \ln x\,dx$

$u = \ln(x) \quad dv = x^3dx$

$\int x^2 \ln x \, dx = \ln(x)\frac{x^3}{3} - \frac{x^3}{9}$

$3\left(\ln(x)\frac{x^3}{3} - \frac{x^3}{9}\right) + C$

$x^3 \ln(x) - \frac{x^3}{3} + C$

$du = \frac{1}{x}dx \quad v = \frac{x^4}{4} \quad uv - \int v\,du$

$\int x^3 \ln x \, dx = \frac{x^4}{4}\ln x - \int \frac{x^4}{4} \cdot \frac{1}{x} dx$

$\frac{x^4}{4}\ln x \quad -\frac{1}{4}\int x^3 dx$

$\frac{x^4}{4}\ln x - \frac{x^4}{16}$

$2\left(\frac{x^4}{4}\ln x - \frac{x^4}{16}\right) + C$

$\frac{x^4}{2}\ln x - \frac{x^4}{8} + C$

9) $\int e^{2x+3}\cos(2x+3)\,dx$

$u = 2x+3$

$du = 2dx = \frac{1}{2}du$

$\frac{1}{2}\int e^u \cos u \, du \qquad I = \int e^u \cos u \, du$

$w = \cos u \quad dw = -\sin u \, du$

$z = e^u \quad dz = e^u du$

$zw - \int w\,dz$

$e^u \cos u - \int e^u(-\sin u)du$

$e^u \cos u + \int e^u \sin u \, du$

$w = \sin u \quad dw = \cos u \, du$

$z = e^u \quad dz = e^u du$

$\int e^u \sin u \, du = e^u \sin u - \int e^u \cos u \, du$

$e^u \sin u - I$

$I = e^u \cos u + e^u \sin u - I$

10) $\int \frac{x^3 - 2}{x^2 + 1} dx$

$\frac{x(x^2+1) - x - 2}{x^2+1}$

$x - \frac{x+2}{x^2+1}$

$\int \frac{x^3-2}{x^2+1}dx = \int x\,dx - \int \frac{x}{x^2+1}dx - 2\int \frac{1}{x^2+1}dx$

$\int x\,dx = \frac{x^2}{2}$

$u = x^2+1 \quad du = 2x\,dx$

$2I = e^u(\cos u + \sin u)$

$I = \frac{e^u}{2}(\cos u + \sin u)$

$\int e^{2x+3}\cos(2x+3)\,dx = \frac{1}{2}\cdot\left[\frac{e^u}{2}(\sin(u)+\cos(u))\right]+C$

$\frac{1}{4}e^u(\sin u + \cos u)+C$

$\frac{1}{4}e^{2x+3}\left(\sin(2x+3)+\cos(2x+3)\right)+c$

$\int \frac{x}{x^2+1}\,dx = \frac{1}{2}\int \frac{2x}{x^2+1}\,dx = \frac{1}{2}\ln(x^2+1)$

$\int \frac{1}{x^2+1}\,dx = \tan^{-1}(x)$

$\frac{x^2}{2} - \frac{1}{2}\ln\left(x^2+1\right) - 2\tan^{-1}x + C$

$f(x) = \frac{1}{\sqrt{2\pi}}e^{-0.5x^2}$

11) $\int_{-1}^{1} f(x)\,dx = \frac{\Delta x}{2}\left[f(x_0) + 2\sum_{i=1}^{7} f(x_i) + f(x_8)\right]$

$\sum_{i=1}^{7} f(x_i) = f\left(\frac{-3}{4}\right) + f\left(\frac{-2}{4}\right) + f\left(-\frac{1}{4}\right) + f(\emptyset) + f\left(\frac{1}{4}\right) + f\left(\frac{2}{4}\right) + f\left(\frac{3}{4}\right)$



Damn.

Shouldn't waited to do this one last

14) $\frac{dy}{dt} = 3t^2(1+y^2) \qquad y(0) = \frac{\pi}{4}$

$$\frac{1}{1+y^2}\,dy = 3t^2\,dt$$

$$\int \frac{1}{1+y^2}\,dy = \int 3t^2\,dt$$

$$\tan^{-1}(y) = t^3 + C$$

$$\tan^{-1}\left(\frac{\pi}{4}\right) = (0)^3 + C$$

$$\tan^{-1}\left(\frac{\pi}{4}\right) = C$$

$$\tan^{-1}(y) = t^3 + \tan^{-1}\left(\frac{\pi}{4}\right)$$

$$y(t) = \tan\left(t^3 + \tan^{-1}\left(\frac{\pi}{4}\right)\right)$$

15) $\dfrac{dy}{dt} = \dfrac{\cos(t)(y^2-1)}{2y^3}$

$$\frac{2y^3}{y^2-1}\,dy = \cos t\,dt$$

$$\int \frac{2y^3}{y^2-1}\,dy = \int \cos t\,dt$$

$2y(y^2-1)+2y$ $\quad \sin t + C$

$$\frac{\quad}{y^2-1}$$

$$\int\left(2y+\frac{2y}{y^2-1}\right)$$

$$y^2+\ln|y^2-1|$$

$$y^2+\ln|y^2-1|=\sin t+c$$

16) $V=120L$

$r_{in}=5\,L/min \quad r_{out}=5L/min$

$c_{in}=2.3\,kg/L \quad c_{out}=\frac{y(t)}{120}\,kg/L$

$y(0)=0.5\,kg$

$$\frac{dy}{dt}=r_{in}(5L/min)(2.3kg/L)-r_{out}(5L/min)\left(\frac{y}{120}kg/L\right)$$

$$11.5kg/min-\frac{y}{24}kg/min$$

$$\frac{dy}{dt}=11.5-\frac{y}{24}$$

$$\frac{dy}{dt}=\frac{276-y}{24}$$

$$\int\frac{1}{276-y}\,dy=\int\frac{1}{24}\,dt$$

$y(0)=0.5$

$y(t)=120$

$0.5=276-Ae^{0} \quad 275.5$

$120=276-275.5e^{-t/24}$

$e^{-t/24}=\frac{156}{275.5}=\frac{312}{551}$

$$-\ln|276-y| = \frac{t}{24} + C$$

$$\ln|276-y| = -\frac{t}{24} - C$$

$$276-y = Ae^{-t/24}$$

$$y(t) = 276 - Ae^{-t/24}$$

$$\frac{-t}{24} = \ln\left(\frac{312}{551}\right)$$

$$t = -24(\ln 312 - \ln 551)$$

calculator

$$t \approx 13.65 \text{ minutes}$$